Preparation for Checkpoint 2 - CITS3403

1. "Decide what application you are going to build. Remember it must satisfy the requirements laid out on the "Information and FAQs" page in this folder."

For our project, we plan to build a custom study platform inspired by Quizlet, fully developed using HTML, CSS, JavaScript, Flask, and SQLite, that allows users to create, study, and share flashcards in an engaging and intuitive way. Users will be able to register, log in, and maintain personal profiles that track their progress and study habits. The platform will support both manual flashcard creation and AI-assisted generation from lecture notes or text input, making it easier to build comprehensive study sets. Flashcards can be studied through multiple modes, including standard flip, quizzes, and timed memory games, with progress tracking to highlight learnt and review-needed cards. Users will also be able to share their sets with others, view public sets, and organise them with tags for easy discovery. A user-friendly dashboard will provide analytics such as total sets created, cards studied, and progress visualisations through charts. The platform will feature responsive design and intuitive navigation, ensuring an effective, engaging, and collaborative study experience while demonstrating our ability to implement a client-server architecture, persist data across sessions, and apply agile project methodologies within a team.

2. Write at least 10 user stories that describe proposed interactions with your system **(see Lecture 1 slides for details).**

Epic A: User Authentication

A1. User registers for an account

- As a student, I want to sign up for an account with an email as a username and password so that I can access personalised flashcard sets and track my study progress.

Acceptance Criteria (AC):

- Given I am on the Sign-Up Page, when I enter a valid email, password, password confirmation, then my user account is created and registered in the database, with redirection to the Login Page.
- Given that I submit invalid details (e.g. mismatched password, duplicate email), when I register for an account, then an error message will be displayed without creating an account.

A2. User logs into the account

- As a registered user, I want to log in with my credentials so that I can access my User Dashboard to access various Flashcard Set Pages, Profile Page, Analytics Page, Search and Setting Page.

AC:

- Given I am on the Login Page, when I enter a valid email and its corresponding password, I am redirected to the User Dashboard.
- Given my Login details are incorrect, when I submit, an error message is shown and I am not logged in.

Epic B: Flashcard Set Creation

B1. User creates Flashcard Set manually

- As a registered user, I want to create a new flashcard set by submitting terms and definitions so that I can build a custom study set to enhance my learning.

AC:

- Given I am on the Create Flashcard Set Page in Dashboard, when I enter a title, description, and at least one term-definition pair, then the set is saved into the Database and displayed in my Dashboard .
- Given I add more than one card using (INSERT SOFTWARE), when I save the page, all cards are correctly linked into the set.

B2. User generates Flashcard Set with AI assistance

- As a registered user, I want to generate flashcards automatically when given a prompt or lecture notes so that I can quickly create subjected sets without defining each card manually.

AC:

- Given I am on the Create Flashcard Set Page and type into the AI bot, when I paste some text or type in a prompt, then Generate, a list of suggested cards appear.
- Given I review or edit these generated cards, then save, the final set is saved into the Database and is visible in my Dashboard [AI incorporation is in backend*].

Epic C: Study Created Flashcards

C1. User studies their sets in flip mode

- As a registered user, I want to study my own flashcard set in flip mode so that I can actively recall information by manually flipping cards.

AC:

- Given I open a set from the Dashboard, then View Flashcard Set Page, when I press Flip mode, then cards display the front and back, changeable by clicking or tapping, with buttons to mark Easy/Medium/Hard.
- Given I mark my cards during the study session, when I finish or exit, my progress is saved (e.g. cards marked Medium/Hard will be **flagged in review mode).

C2. User takes a quiz on flashcard set

- As a registered user, I want to take an interactive quiz based on my flashcard set so that I can test my knowledge in a multiple-choice/short answer/both format.

AC:

- Given I am viewing my set, when I choose Review (Quiz) mode, questions are randomly generated from the cards. I select/input answers, and immediate feedback is shown and information is gathered.
- When I have finished my quiz, a score summary appears and my progress is updated in the database.

C3. User plays a Timed memory game

- As a registered user, I want to play a timed memory game with my flashcards so that I am fully engaged and I can recall information quicker.

AC:

- Given I select Timed Memory Game mode on my set, when the game starts, a timer begins and cards are presented (match terms to definition) and points are awarded accordingly.
- Given the timer runs out or I finish on time, then the results (score, time taken) are displayed and updated to my progress.

Epic D: Sharing & Discovery

D1. User shares a set publicly

- As a registered user, I want to make my flashcard set public or shareable so that other students can study with my materials.

AC:

- Given I own a set (on View Flashcard Set Page), when I press 'Make Public' and save, the set appears on the Public Sets Page for others to view/study.
- Given a set is public, when another user browses public sets, then they can open and study it without editing.

D2. User browses and searched public sets

- As a user (regardless of registered or not), I want to search and browse public flashcard sets by keywords so that I can find relevant study materials created by others.(Hoang)

AC:

- Given I am on the Search Page or Public Sets Page, when I enter a relevant keyword, matching public sets are displayed with the **Title, Creator, Card Count and Preview.**
- Given I click a suggested result, when I open it, I am taken to the View Flashcard Set page in study-mode.

Epic E: Analytics Page

E1. User views Personal Study Analytics

- As a registered user, I want to view my study progress and statistics on a summary page so that I can track my progress, see which cards need reviewing/are mastered, and be engaged.

AC:

- Given I am on the Analytics Page, from Dashboard, when loaded, visual charts and figures are displayed (e.g. total sets created, total cards studied, study streak, % correct, Needs Review)
- Given progress data is tracked in study sessions, when I view analytics, the data is up-to-date and accurate [Insert software]

3. Create a list of the main pages your website will have (e.g. for a movie rating review site: signup page, login page, view movies page, add new movie page, search movie pages etc.)

3.1. Landing/Home Page: Introduction to the platform, featured public flashcard sets, and a call to action to sign up or log in.
3.2. Signup Page: Create a new user account.
3.3. Login Page: Access an existing account.
3.4. User Dashboard: Overview of the user's flashcard sets, progress stats, and recent activity.
3.5. Create Flashcard Set Page: Form to create a new flashcard set manually or via AI-assisted generation.
3.6. View Flashcard Set Page: Display a flashcard set with study options such as flip cards, quiz mode, and timed games.
3.7. **Shared/Public Sets Page: Browse and search sets shared by other users.
3.8. Profile Page: Edit personal details, view owned sets, and track study achievements.
3.9. Search Page: Search for flashcards or sets by keyword or tags.
3.10. Analytics Page: Visualisations of study progress, such as number of cards studied, sets completed, and streaks.
3.11. Settings Page: Account settings, notifications, and optional preferences.
3.12. Error/404 Page: Friendly error page for invalid URLs or actions.

```
SITEMAP


Landing/Home Page
│
├── Signup Page
├── Login Page
│
├── User Dashboard
│    ├── Create Flashcard Set Page
│    │    └── View Flashcard Set Page (study modes: flip, quiz, timed games)
│    ├── View Flashcard Set Page (study own sets)
│    ├── Shared/Public Sets Page
│    │    └── View Flashcard Set Page (study others' sets)
│    ├── Profile Page
│    ├── Analytics Page
│    ├── Search Page
│    │    └── View Flashcard Set Page (study searched sets)
│    └── Settings Page
│
```

```
└─ Error/404 Page
```

4. Decide which CSS Framework you are going to use.

For our AI flashcard study platform, we suggest using Tailwind CSS, as it allows us to create a clean, modern, and highly customisable dashboard. Tailwind makes it easy to build a user-friendly interface with a responsive layout and visually appealing pages, while also integrating smoothly with Flask templates for our backend. This framework will enable us to design an engaging and intuitive platform that meets the project's usability and aesthetic goals.

5. (*Optional but strongly encouraged, especially for groups whose meeting is in the later weeks*) Start creating some of your pages with HTML and CSS. The page does not need to be interactive (e.g. no need for buttons to work) but it should be sufficient for a prospective client to understand how the website will function. You should commit all your files to GitHub.

Can we have a couple of designs ready? (No need for HTML / CSS) Instead, create a prototype on figma.com

hungvu120406@gmail.com

```
Signup Page
├─ Login Page

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│    ├─ Profile Page
│    ├─ Analytics Page
│    ├─ Search Page
│    │    └─ View Flashcard Set Page (study searched sets) == others' sets as it's
public
│    └─ Settings Page
```

silvictory2005@gmail.com

```
│    ├─ Create Flashcard Set Page
│    │    └─ View Flashcard Set Page (study modes: flip, quiz timed games)
│    ├─ View Flashcard Set Page (study own sets)
│    ├─ Shared/Public Sets Page
```

Reference

Introduction

In Checkpoint 2 you will demonstrate to your facilitator that you have a concrete idea for your web-application and that you have an initial plan for what functionality your application will provide and what it will look like.

Preparation for the Checkpoint Meeting

Before the meeting occurs, together as a group you will need to:

- Decide what application you are going to build. Remember it must satisfy the requirements laid out on the "Information and FAQs" page in this folder.
- Write at least 10 user stories that describe proposed interactions with your system (see Lecture 1 slides for details).
- Create a list of the main pages your website will have (e.g. for a movie rating review site: signup page, login page, view movies page, add new movie page, search movie pages etc.)
- Decide which CSS Framework you are going to use.
- (*Optional but strongly encouraged, especially for groups whose meeting is in the later weeks*) Start creating some of your pages with HTML and CSS. The page does not need to be interactive (e.g. no need for buttons to work) but it should be sufficient for a prospective client to understand how the website will function. You should commit all your files to GitHub.

Note: do not start implementing the back-end of your website, i.e. the flask server, until after we have covered it in class after the mid-semester break.

During the Checkpoint Meeting

All members of the team must attend. During the 20 minute meeting:

- You will describe your proposed application to the facilitator.
- You will show them your planning documents and any HTML pages you have made.
- The facilitator will provide feedback.
- The facilitator will ask briefly about your team organisation, e.g. meeting schedule.

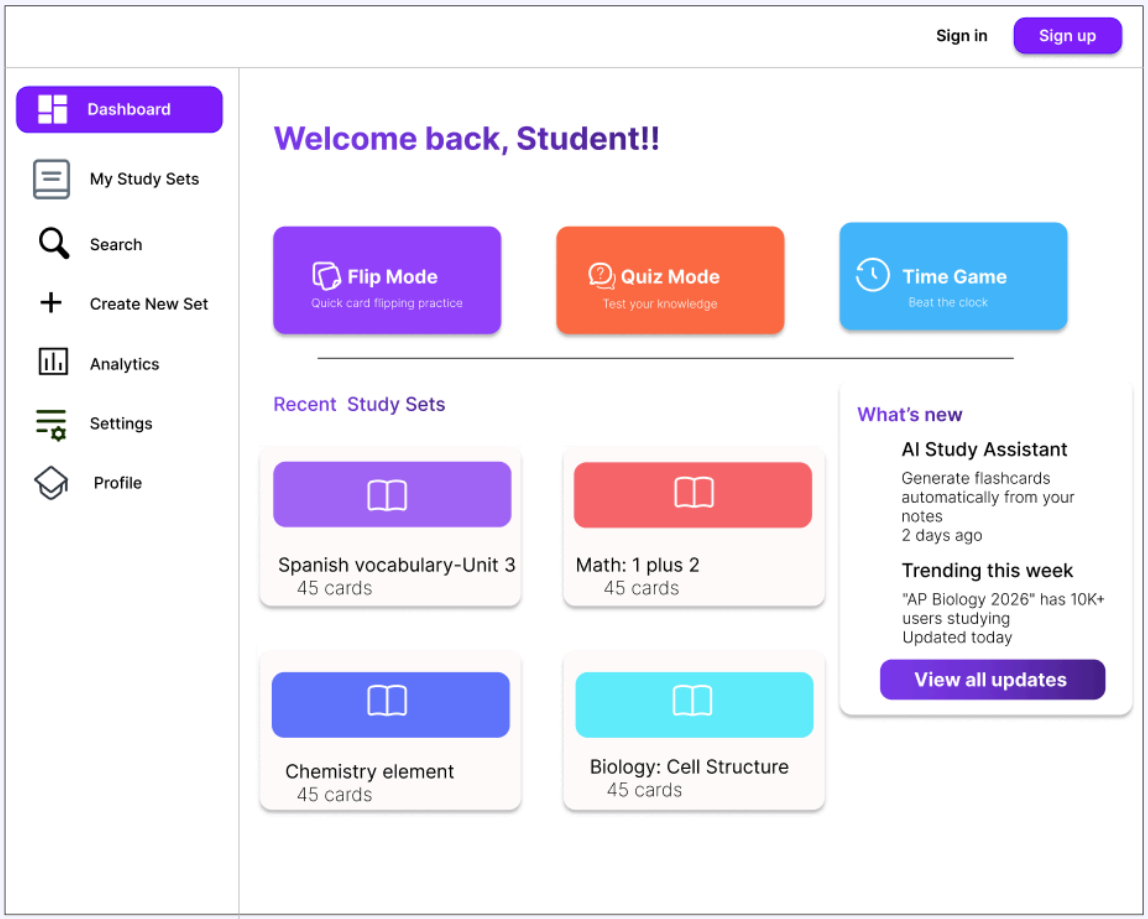
Checkpoint 2: Application Planning (Source LMS CITS3403)

Roles for Mid Sem/Week 7 (not finalised):

Silvi - Navigation
Casey - Flashcards
Ani - AI
Vu - Quiz mode/time/charts

FIGMA

Desktop - 1


Sign in
Sign up
Dashboard
My Study Sets
Search
Create New Set
Analytics
Settings
Profile
Welcome back, Student!!
Flip Mode
Quick card flipping practice
Quiz Mode
Test your knowledge
Time Game
Beat the clock
Recent Study Sets
Spanish vocabulary-Unit 3
45 cards
Math: 1 plus 2
45 cards
Chemistry element
45 cards
Biology: Cell Structure
45 cards
What's new
AI Study Assistant
Generate flashcards automatically from your notes
2 days ago
Trending this week
"AP Biology 2026" has 10K+ users studying
Updated today
View all updates

Desktop - 1

Welcome Back

Login with Gmail

Login with Facebook

Login with Apple

Or

Email

Enter your email or username

Password

Forget password ?

Enter your password

Log in

Dont have an account ? Sign up

Desktop - 1

Create your account

Start your learning yourney today

Full name

Email Address

Password

Confirm Password

I agree to the Terms and Conditions and Privacy Policy

Create Account

Or continue with

Google

Github

Already have an account? Sign in

Desktop - 1

Dashboard

My Study Sets

Search

Create New Set

Analytics

Settings

Profile

Lynn Ta

Medical student

1,204
Cards study

12 Days
Study Streak

4
Achievement

My Study Set

Medical Terminology
120 terms

Human Anatomy Basics
59 terms

Pharmacology Fundamentals
45 terms

Lynn Ta

Dashboard

Bo...

My Study Sets

Search

Create New Set

li...

Analytics

lis...

Settings

Profile

Search Public Study Set

Find your perfect flashcards topic for your learning

Search for study sets...

Biology: Cell Structure and Function

45 Terms

Chemistry: Periodic Table Elements

57 Terms

Spanish Vocabulary - Beginner Level

100 Terms

Mathematics: Calculus Formulas

72 Terms

Lynn Ta

Dashboard

My Study Sets

Search

Create New Set

Analytics

Settings

Profile

Settings

Manage your account

Account

Username

abcxyz....

Email

a@gmail.com

Notifications

- [ ] Study reminder
- [ ] Daily goal notifications
- [ ] Friend activities
- [ ] New study set updates

Language

English

Desktop - 1

Lynn Ta

Dashboard
My Study Sets
Search
Create New Set
Analytics
Settings
Profile

Analytics

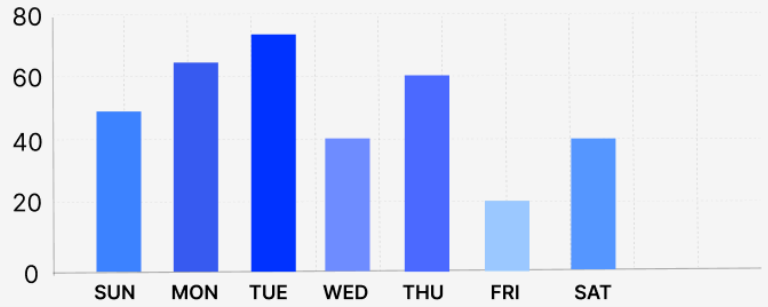
Your weekly flashcard study process

376
Cards studied

89%
Accuracy

12
Days Streak

This Week



Frame 1

Lynn Ta

Dashboard
My Study Sets
Search
Create New Set
Analytics
Settings
Profile

My Study Sets

Create, manage, and study your flashcard sets

History
Important events in world history
2 cards
Study

Programming
Essential Python concepts
2 cards
Study

Python Logic
Covers range of topic in Python
1 card
Study

Science
Fundamental concepts of Physics and Biology
10 cards
Study

Back to Dashboard

Edit Study Set

Build your own custom flashcard set

Set Title *

Programming

Description

Learn about for loops,while loop, and syntax.

Question 1

Question (Term) *

Why do we use a for loop in Python?

Correct Answer (Definition) *

To repeat a block of code for each item in a sequence

Multiple Choice Options for Quiz (Please include the correct answer as well)

To repeat a block of code for each item

To define a new function

To stop a program immediately

To handle errors in a program

Question 2

Question (Term) *

Why do we use a while loop in Python?

Correct Answer (Definition) *

To repeat a block of code as long as a condition is true

Multiple Choice Options for Quiz (Please include the correct answer as well)

To stop the program execution

To define a new function

To repeat a block of code as long as a condition is true

To store multiple values in a list

+ Add question

Save Changes

Frame 1

← Back to Dashboard

Programming

Essential Python concepts

Covers range of topic in Python

1 card

Card Difficulty Breakdown

Easy	Medium	Hard
1	0	0

Lynn Ta

Dashboard
My Study Sets
Search
Create New Set
Analytics
Settings
Profile

Back to Dashboard

Create New Study Set

Build your own custom flashcard set

Set Title *

e.g., Python Logic - Week 1

Description

Brief description of what this set covers...

AI Flashcard Generator

Paste your notes or type a prompt...

e.g. paste your notes such as "Python does not require data type declaration unlike C which will throw an error if undeclared" or type a prompt such as "Make me 10 flashcards about Python syntax"

Generate Flashcards

Question 1

Question (Term) *

Enter your question...

Correct Answer (Definition) *

Enter the correct answer...

Multiple Choice Options for Quiz (Please include the correct answer as well)

Option 1 | Option 3

Option 2 | Option 4

+ Add question | Create Study Set

Lynn Ta

Dashboard
My Study Sets
Search
Create New Set
Analytics
Settings
Profile

Back to Dashboard

Create New Study Set

Build your own custom flashcard set

Set Title *

Python Logic

Description

Covers range of topic in Python

AI Flashcard Generator

Make me a flashcard about Python logic

e.g. paste your notes such as "Python does not require data type declaration unlike C which will throw an error if undeclared" or type a prompt such as "Make me 10 flashcards about Python syntax"

Generate Flashcards

Generated a flashcard

Card 1

Add to Set

Question: What are the three logical operators in Python?

Answer: and, or, and not. Used to combine or reverse boolean conditions.

Question 1

Question (Term) *

Enter your question...

Correct Answer (Definition) *

Enter the correct answer...

Multiple Choice Options for Quiz (Please include the correct answer as well)

Option 1

Option 3

Option 2

Option 4

+ Add question

Create Study Set

Frame 3 (When add to set, no mcq available as it only set for flashcard)

Lynn Ta

Dashboard
My Study Sets
Search
Create New Set
Analytics
Settings
Profile

Back to Dashboard

Create New Study Set

Build your own custom flashcard set

Set Title *

Python Logic

Description

Covers range of topic in Python

AI Flashcard Generator

Paste your notes or type a prompt...

e.g. paste your notes such as "Python does not require data type declaration unlike C which will throw an error if undeclared" or type a prompt such as "Make me 10 flashcards about Python syntax"

Generate Flashcards

Question 1

Question (Term) *

What are the three logical operators in Python?

Correct Answer (Definition) *

and, or, and not. Used to combine or reverse boolean conditions.

Multiple Choice Options for Quiz (Please include the correct answer as well)

Option 1

Option 3

Option 2

Option 4

+ Add question

Create Study Set

Frame 1

Back to Dashboard

Review Progress

Python Logic

Choose your study mode

Flip Mode

Study at your own pace with interactive flashcards

Quiz Mode

Test your knowledge with multiple choice questions

Time Game

Challenge yourself with a timed quiz

Frame 1

Back to Dashboard

Programming

Card 1 of 2

Question

Why do we use a for loop in Python?

Click to flip

1 2

Next >

Frame 2

Back to Dashboard

Programming

Card 1 of 2

Answer

To repeat a block of code for each item in a sequence

Click to flip

How well did you know about this?

Easy Medium Hard

1 2

Next >

Frame 3

Back to Dashboard

Programming

Card 2 of 2

Question

Why do we a while loop in Python?

Click to flip

1 2

Next >

Frame 4

Back to Dashboard

Programming

Card 2 of 2

Answer

To repeat a block of code as long as a condition is true

Click to flip

How well did you know about this?

Easy Medium Hard

1 2

Next >

Frame 1

Back to Dashboard

Programming

Question 1 of 2

Why do we use a for loop in Python?

- To repeat a block of code for each item ✓
- To stop a program immediately
- To define a new function
- To handle errrors in a new program

1 2

Next >

Frame 2

Back to Dashboard

Programming

Question 2 of 2

Why do we use a while loop in Python?

- To stop the program execution ✕
- To repeat a block of code as long as a condition is true
- To define a new function
- To store multiple values in a list

1 2

View Result

Frame 3

← Back to Dashboard

Quiz Result

50%	1	1
Score	Correct	Incorrect

Detailed Results

Status	Question	Your Answer	Correct Answer
Correct	Why do we use a for loop in Python?	To repeat a block of code for each item	To repeat a block of code for each item
Incorrect	Why do we use a while loop in Python?	To stop the program execution	To repeat a block of code as long as a condition is true

Play Again Choose Another Set

Frame 1

Back to Dashboard

Programming

Question 1 of 2

60s

Why do we use a for loop in Python?

- To repeat a block of code for each item ✓
- To stop a program immediately
- To define a new function
- To handle errrors in a new program

Score: 1/2

1 **2**

Next >

Frame 3

Frame 2

Back to Dashboard

Programming

Question 2 of 2

5s

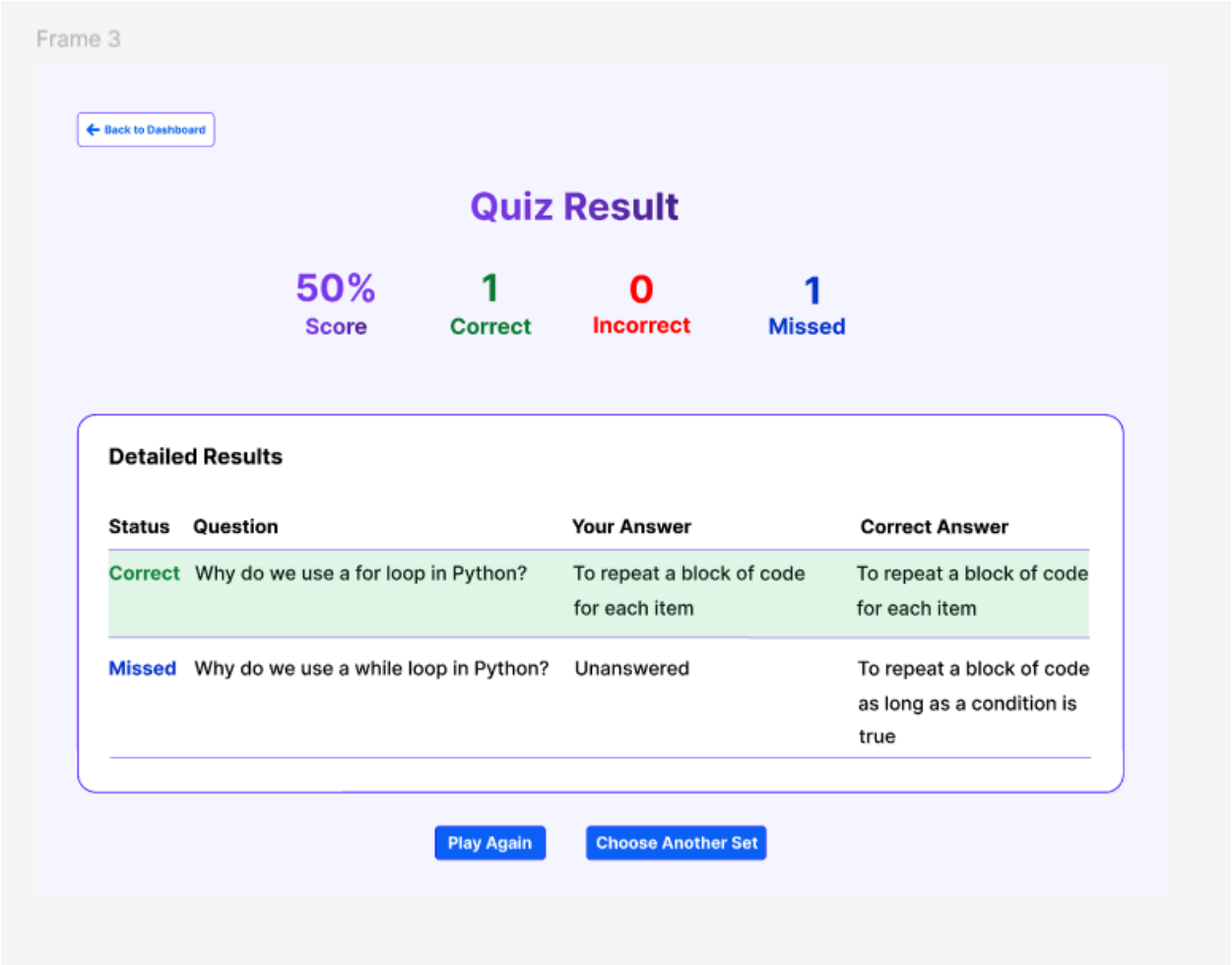
Why do we use a while loop in Python?

- To stop the program execution
- To repeat a block of code as long as a condition is true
- To define a new function
- To store multiple values in a list

Score: 1/2

1 2

View Result



https://www.figma.com/design/dffUpEFcqruTQdvCVOlo3L/Agile?node-id=78-5&p=f&t=oBfK1foykKzhVwfs-0